Economics Letters 169 (2018) 35–37

Contents lists available at ScienceDirect

Economics Letters

journal homepage: www.elsevier.com/locate/ecolet

A characterization of the Esteban–Ray polarization measures

Yoko Kawada, Yuta Nakamura, Keita Sunada *

Graduate School of Economics, Keio University, 2-15-45 Mita, Minato-ku, Tokyo, Japan

HIGHLIGHTS

- We present a counterexample to Esteban–Ray 1994 polarization measure theorem.
- We strengthen their axiom so that their characterization result is reestablished.
- Our modified axiom does not change the spirit of the original axiom.

ARTICLE INFO

Article history:
Received 20 February 2018
Received in revised form 10 May 2018
Accepted 10 May 2018

JEL classification:
D31
D63

Keywords:
Polarization
Esteban–Ray polarization measures

ABSTRACT

Esteban and Ray (1994) formalized the idea of polarization and developed a theory for its measurement. In their main theorem, they claimed that a class of polarization measures, called the *Esteban–Ray measures*, is characterized by a set of axioms that capture the idea of polarization. However, in this study, we show that the claim does not hold by presenting a counterexample. We amend their main theorem by strengthening the first axiom.

© 2018 Elsevier B.V. All rights reserved.

1. Introduction

A seminal work by Esteban and Ray (1994) (henceforth "ER") formalized the idea of *polarization* and developed a theory for its measurement. In their main theorem (ER, Theorem 1), they claimed that a class of allowable polarization measures, called the *Esteban–Ray measures*, is characterized by a set of axioms that capture the idea of polarization. However, by presenting a counterexample, we show that this claim does not hold. Further, we strengthen their "Axiom 1" so that the resulting characterization is reestablished.

The rest of this study is organized as follows. Section 2 introduces definitions and axioms. Section 3 presents our results. Proofs are relegated to Supplementary Appendix.

2. Model and axioms

Our model follows that of ER. Let $\mathbb{R}$ be the set of *attributes* (a basic perceptual variable is the natural logarithm of income). We consider population distributions on $\mathbb{R}$ with finite supports. That is, a distribution is denoted by a pair of n-dimensional vectors $(\boldsymbol{\pi}, \boldsymbol{y}) = ((\pi_1, \ldots, \pi_n), (y_1, \ldots, y_n)) \in \mathbb{R}^n_{++} \times \mathbb{R}^n$ for some $n \in \mathbb{N}$, where π_i is the population of individuals with attribute y_i and $y_i \neq y_j$ for distinct $i, j \in \{1, \ldots, n\}$. Let

$$\mathscr{D} \equiv \bigcup_{n=1}^{\infty} \mathbb{R}^n_{++} \times \left\{ \boldsymbol{y} \in \mathbb{R}^n : \ y_i \neq y_j \text{ for all distinct } i, j \in \{1, \ldots, n\} \right\}$$

be the set of distributions. A *polarization measure* is a function $P : \mathscr{D} \to \mathbb{R}_+$ that maps each distribution $(\boldsymbol{\pi}, \boldsymbol{y}) \in \mathscr{D}$ to a non-negative real number $P(\boldsymbol{\pi}, \boldsymbol{y}) \in \mathbb{R}_+$. ER's analysis focused on polarization measures that take the following functional form:

$$P\left((\boldsymbol{\pi}, \boldsymbol{y})\right) = \sum_{i=1}^{n} \sum_{j=1}^{n} \pi_i \pi_j \theta\left(\pi_i, \left|y_i - y_j\right|\right), \tag{1}$$

where θ is a function $\mathbb{R}^2_+ \to \mathbb{R}$ such that $\theta(\cdot, \cdot)$ is strictly increasing in the second argument (distance), continuous in each argument, $\theta(0, \cdot) = 0$, $\theta(\cdot, 0) = 0$, and $\theta(\pi_i, \delta) > 0$ for all $\pi_i > 0$ and $\delta > 0$.[1] An interpretation and background of this form is discussed in ER.

* Corresponding author.
E-mail address: sunada@keio.jp (K. Sunada).

https://doi.org/10.1016/j.econlet.2018.05.011
0165-1765/© 2018 Elsevier B.V. All rights reserved.

[1] The assumption $\theta(0, \cdot) = 0$ is not imposed by ER. However, this assumption is necessary to deduce Equation (6) in their proof of Theorem 1.

ER proposed three axioms that capture the idea of polarization. Figs. 1–3 show an illustration of each axiom. Axiom 1 says that when a large mass exists at attribute 0, unifying two close masses increases polarization.

Axiom 1. *For any $p > 0$ and any $x > 0$, there exist $\varepsilon > 0$ and $\mu > 0$ such that, for any $y > x$ and any $q < p$ with $y - x < \varepsilon$ and $0 < q < \mu p$,*

$$P\left(((p, q, q), (0, x, y))\right) < P\left(\left((p, 2q), \left(0, \frac{x+y}{2}\right)\right)\right).$$

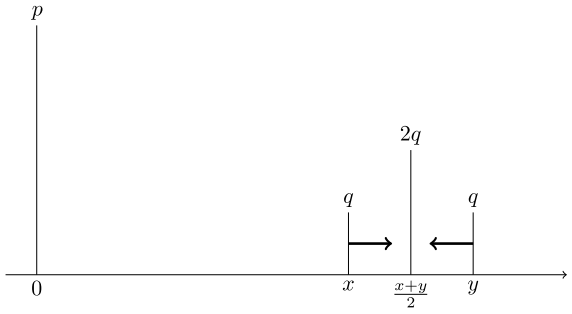


Fig. 1. An illustration of Axiom 1.

Axiom 2 requires that when an intermediate mass gets closer to the right extreme mass, polarization increases.

Axiom 2. *For any $p, q, r > 0$ with $p > r$, any $x, y > 0$ with $|y - x| < x < y$, and any $\Delta \in (0, y - x)$,*

$$P\left((p, q, r), (0, x, y)\right) < P\left((p, q, r), (0, x + \Delta, y)\right).$$

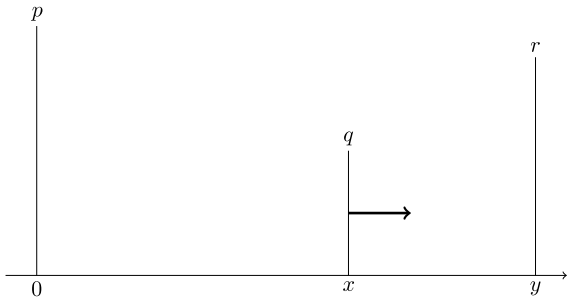


Fig. 2. An illustration of Axiom 2.

Axiom 3 requires that if the population of an intermediate mass decreases, and if the population of left and right extreme masses increase equally, then polarization increases.

Axiom 3. *For any $p, q > 0$, any $x, y > 0$ with $x = y - x$, and any $\Delta \in (0, q/2)$,*

$$P\left((p, q, p), (0, x, y)\right) < P\left((p + \Delta, q - 2\Delta, p + \Delta), (0, x, y)\right).$$

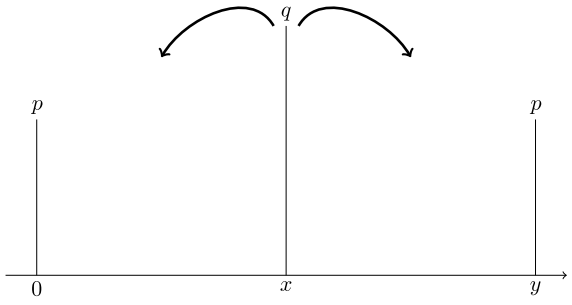


Fig. 3. An illustration of Axiom 3.

Condition H is a homotheticity property requiring any bilateral comparison to be invariant to the scale of population.

Condition H. *For any $(\boldsymbol{\pi}, \boldsymbol{y}), (\boldsymbol{\pi}', \boldsymbol{y}') \in \mathscr{D}$ and $\lambda > 0$, if $P(\boldsymbol{\pi}, \boldsymbol{y}) \geq P(\boldsymbol{\pi}', \boldsymbol{y}')$, then $P(\lambda\boldsymbol{\pi}, \boldsymbol{y}) \geq P(\lambda\boldsymbol{\pi}', \boldsymbol{y}')$.*

3. Main results

3.1. Counterexamples

ER claimed that the class of the Esteban–Ray measures is characterized by Axioms 1–3 and Condition H.

Claim 1 (*ER, Theorem 1*)**.** *A polarization measure P^* of the family defined in (1) satisfies Axioms 1, 2, and 3, and Condition H if and only if it is of the form*

$$P^*(\boldsymbol{\pi}, \boldsymbol{y}) = K \sum_{i=1}^{n} \sum_{j=1}^{n} \pi_i^{1+\alpha} \pi_j \left|y_i - y_j\right| \tag{2}$$

for some constants $K > 0$ and $\alpha \in (0, \alpha^*]$ where $\alpha^* \simeq 1.6$.[2]

We show that Claim 1 does not hold because Axiom 1 is too weak to characterize the class of the Esteban–Ray measures. In their proof of Claim 1, ER showed that Axiom 1 and the continuity of $\theta(\cdot, \cdot)$ in Eq. (1) imply that $\theta(\pi_i, \cdot)$ must be *locally concave* with respect to the distance; that is, for each $x > 0$, there exists $\varepsilon > 0$ such that $\theta(\pi_i, \cdot)$ is concave on a half-open interval $[x, x + \varepsilon)$. Then, they claim that this *local concavity* of θ implies that $\theta(\pi_i, \cdot)$ must be *concave* on $\mathbb{R}_+$. However, this claim is not correct. To see this, fix any $c \in \mathbb{R}_{++}$ and let $\hat{f} : \mathbb{R}_+ \to \mathbb{R}_+$ be such that for each $\delta \in \mathbb{R}_+$,

$$\hat{f}(\delta) = \begin{cases} K\delta & \text{if } \delta < c, \\ K'\delta - (K' - K)c & \text{if } \delta \geq c, \end{cases}$$

where $0 < K < K'$. Then, a convex piecewise linear function $\theta(\pi_i, \delta) = \pi_i^{\alpha} \hat{f}(\delta)$ is not *concave* on $\mathbb{R}_+$, but simply satisfies the *local concavity*.[3] Therefore, Axiom 1 cannot exclude this convex piecewise linear function. In fact, a polarization measure with convex piecewise linear function $\theta(\pi_i, \delta) = \pi_i^{\alpha} \hat{f}(\delta)$ satisfies Axioms 1–3 and Condition H, and hence "Claim 1" does not hold.

Proposition 1 (*Counterexample to Claim 1*)**.** *Let $\hat{P} : \mathscr{D} \to \mathbb{R}_+$ be such that*

$$\hat{P}(\boldsymbol{\pi}, \boldsymbol{y}) = \sum_{i=1}^{n} \sum_{j=1}^{n} \pi_i^{1+\alpha} \pi_j \hat{f}(|y_i - y_j|), \tag{3}$$

where $\alpha \in (0, \alpha^*]$. Then, $\hat{P}$ satisfies Axioms 1, 2, and 3, and Condition H, but does not take the form of (2).

Proof. See Supplementary Appendix A. □

We have two remarks on this proposition; (i) our counterexample function $\hat{P}$ and an Esteban–Ray measure generate different orderings; (ii) a set of counterexamples is *dense* in a set of polarization measures with standard properties. These discussions are relegated to Supplementary Appendix B.

[2] For the definition of α^*, see ER's equation (2) and subsequent arguments on page 833.

[3] To confirm this, consider any $x > 0$. If $x \geq c$, then for any $\varepsilon > 0$, $\theta(\pi_i, \cdot)$ is concave on the half-open interval $[x, x + \varepsilon)$. Conversely, if $x < c$, then by letting $\varepsilon = \frac{1}{2}(c - x) > 0$, $\theta(\pi_i, \cdot)$ becomes concave on the half-open interval $[x, x + \varepsilon)$.

3.2. Modification

We provide a modified axiom that excludes the convex piecewise linear functions, and amend a characterization of the Esteban–Ray measures. For any $x > 0$ and $\varepsilon > 0$, let $B(x, \varepsilon) \equiv \{z \in \mathbb{R}_+ : |x - z| < \varepsilon\}$.

Axiom 1′. *For any $p > 0$ and any $x > 0$, there exist $\varepsilon > 0$ and $\mu > 0$ such that for any $a, b \in B(x, \varepsilon)$ and $q < p$ with $0 < q < \mu p$,*

$$P\left(((p, q, q), (0, a, b))\right) < P\left(\left((p, 2q), \left(0, \frac{a+b}{2}\right)\right)\right).$$

In contrast to Axiom 1, Axiom 1′ implies that $\theta(\pi_i, \cdot)$ must satisfy the following stronger version of *local concavity* with respect to the second argument: for each $x > 0$, there exists $\varepsilon > 0$ such that $\theta(p, \cdot)$ is concave on an open interval $(x - \varepsilon, x + \varepsilon)$. Since convex piecewise linear function $\theta(\pi_i, \delta) = \pi_i^{\alpha} \hat{f}(\delta)$ with kink point c is not concave on any open interval $(c - \varepsilon, c + \varepsilon)$ with $\varepsilon > 0$, θ does not satisfy this *strong local concavity*. This is why Axiom 1′ can exclude the convex piecewise linear functions.

Though Axiom 1′ and Axiom 1 are mathematically quite different, Axiom 1′ has almost the same interpretation as the original compelling axiom. In this sense, this modification does not change the spirit of the original axiom.[4] Now we can restore an axiomatic foundation of the Esteban–Ray polarization measures.

Proposition 2. *A polarization measure P^* of the family defined in (1) satisfies Axioms 1′, 2, and 3, and Condition H if and only if it is of the form*

$$P^*(\boldsymbol{\pi}, \boldsymbol{y}) = K \sum_{i=1}^{n} \sum_{j=1}^{n} \pi_i^{1+\alpha} \pi_j \left|y_i - y_j\right| \tag{2}$$

for some constants $K > 0$ and $\alpha \in (0, \alpha^*]$ where $\alpha^* \simeq 1.6$.

Proof. See Supplementary Appendix C. □

Acknowledgments

The authors would like to thank Toyotaka Sakai for his insightful advice and discussions. We are also grateful to Toru Hokari, Masatoshi Tsumagari, and Noriaki Okamoto for their helpful discussions. We specially thank Joan-María Esteban and Debraj Ray for their kind suggestions. Nakamura's research is financially supported by the Grant-in-Aid for JSPS Fellows (16J00342).

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.econlet.2018.05.011.

References

Esteban, J., Ray, D., 1994. On the measurement of polarization. Econometrica 62 (4), 819–851. http://dx.doi.org/10.2307/2951734.

[4] Assuming differentiability of measures is one way to exclude convex piecewise linear functions. However, differentiability is irrelevant to the original Axiom 1. Moreover, we cannot find a normative reason for adopting differentiability as an axiom in this context. For these reasons, we do not assume differentiability to characterize the Esteban–Ray measures.